Kernix 0.7.2 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

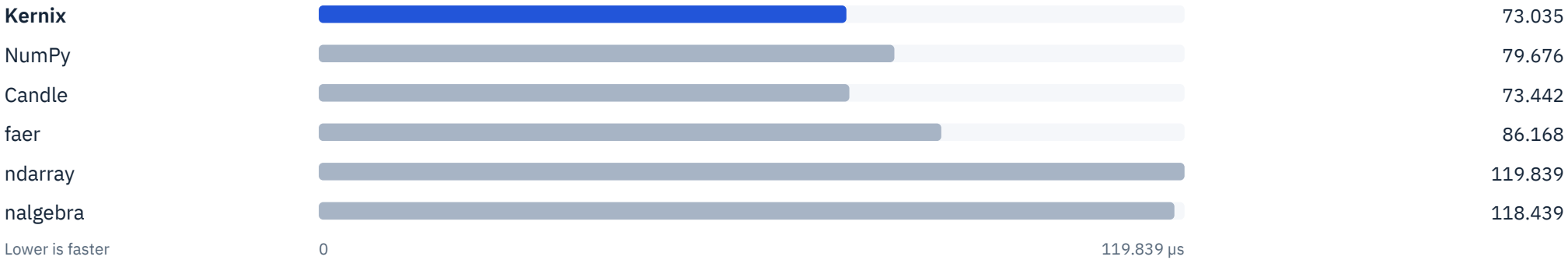
19 workloads

80 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

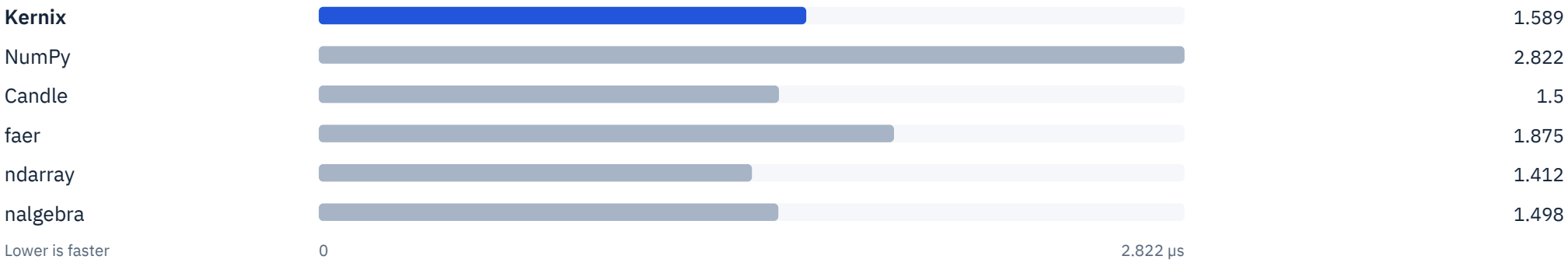
Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

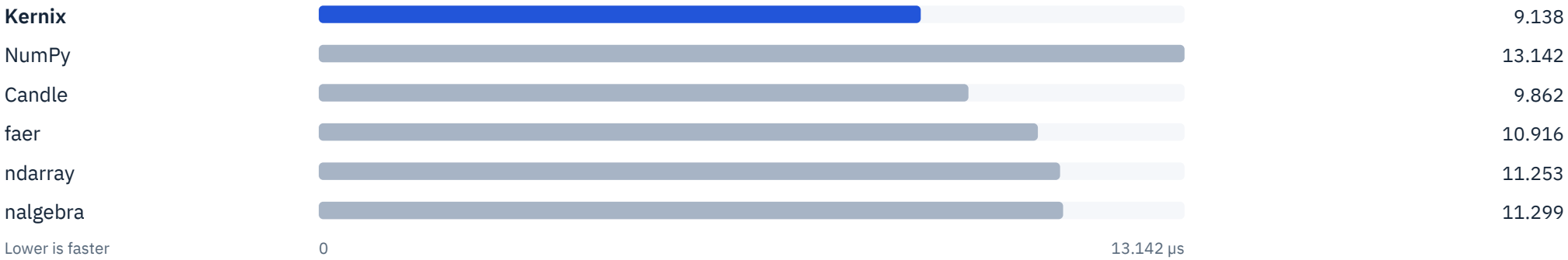
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



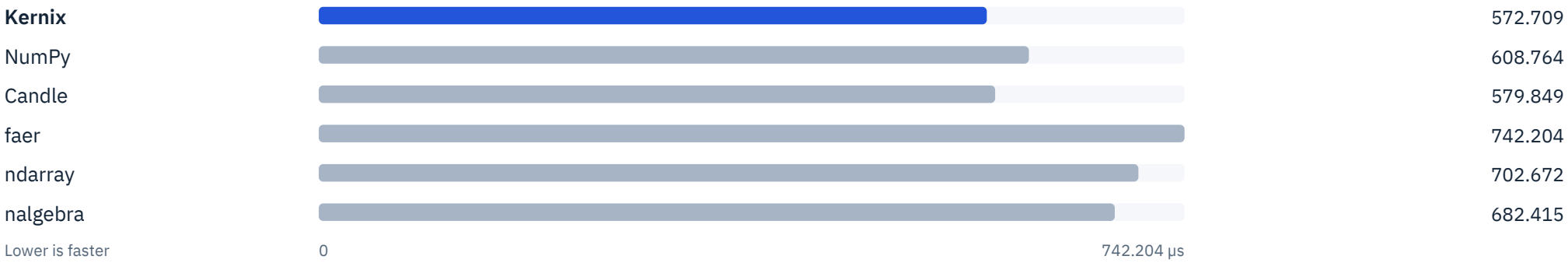
Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

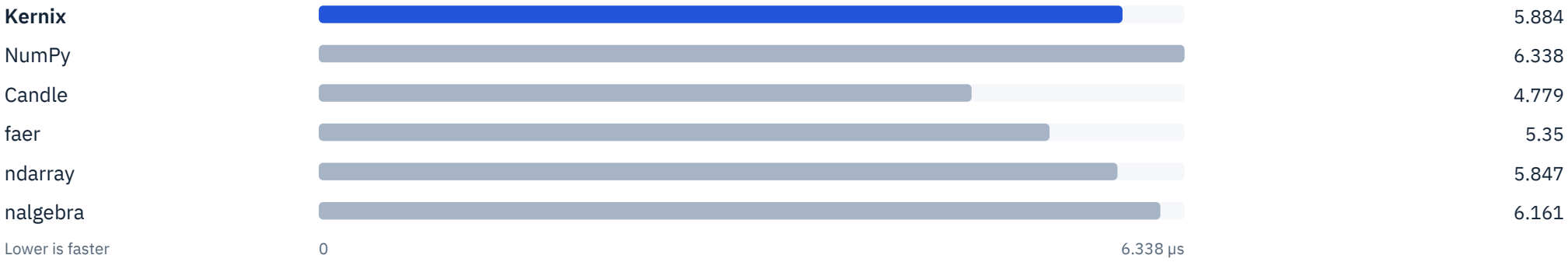
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



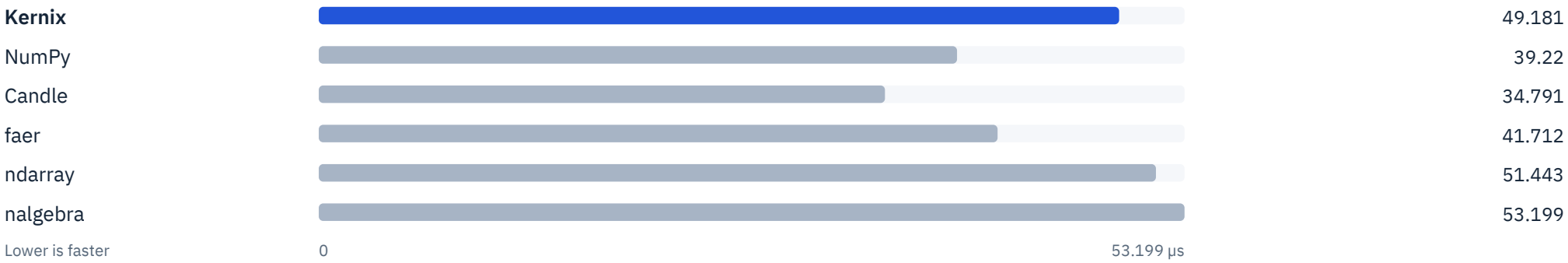
Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

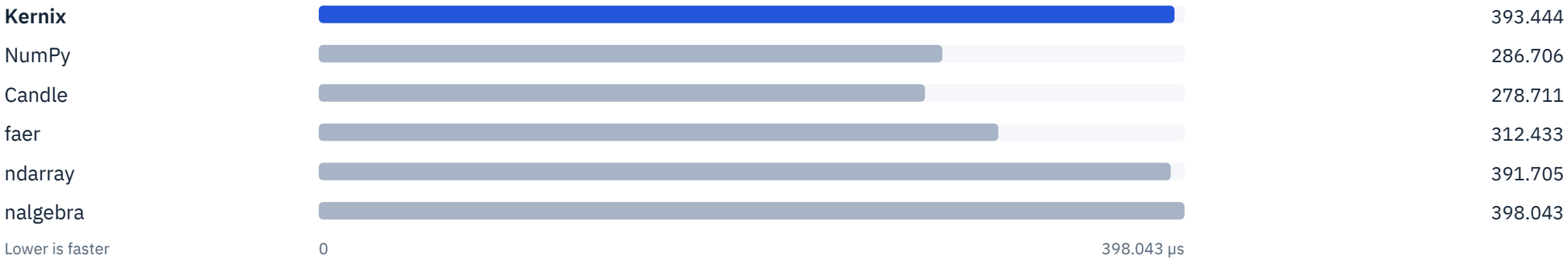
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

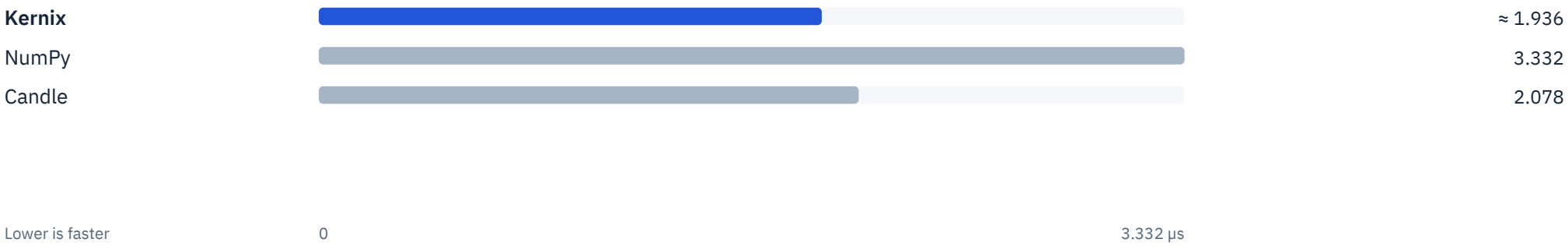


Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

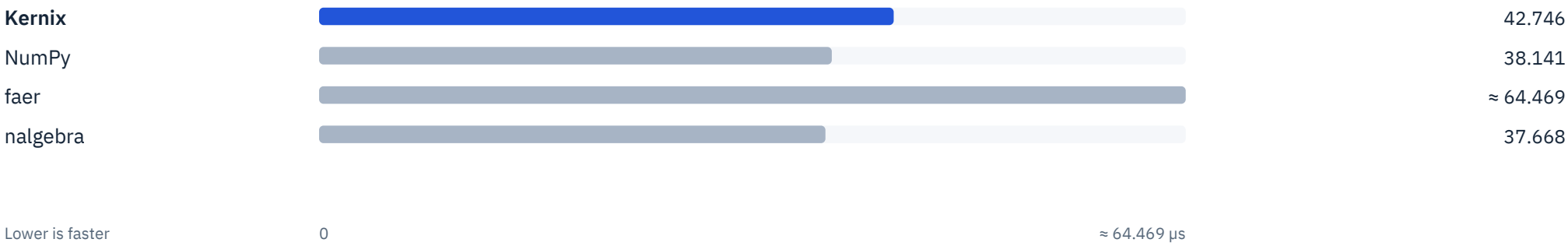


Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output

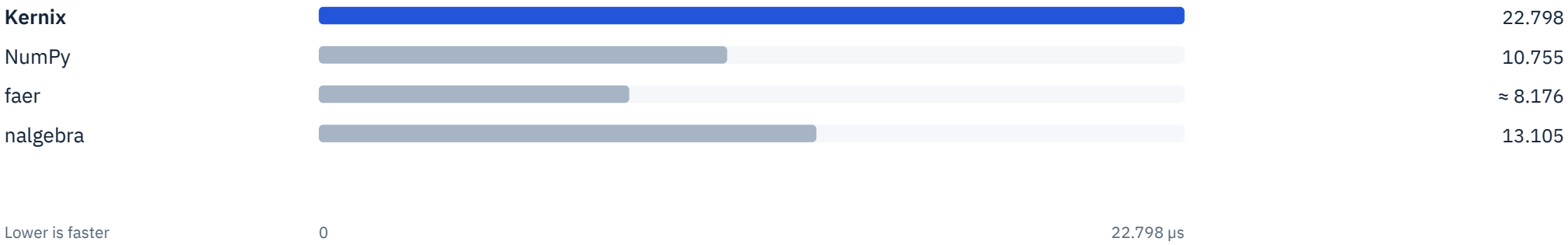


Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

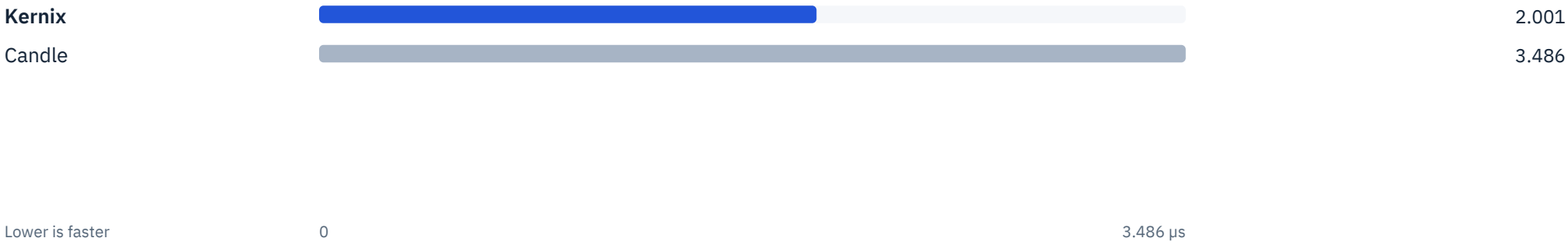
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

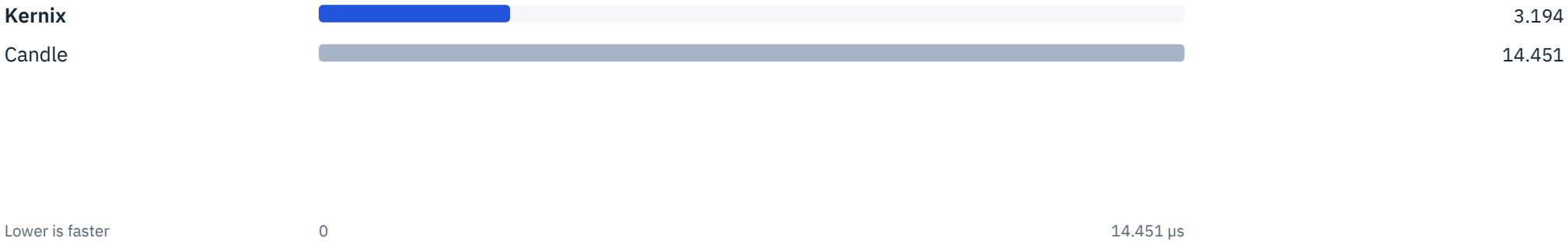
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

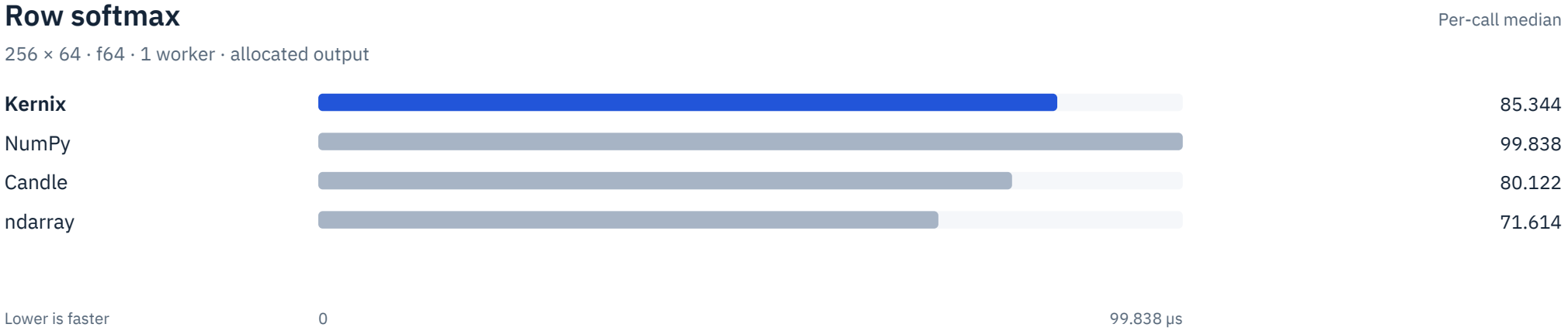


Kernix 0.7.2 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

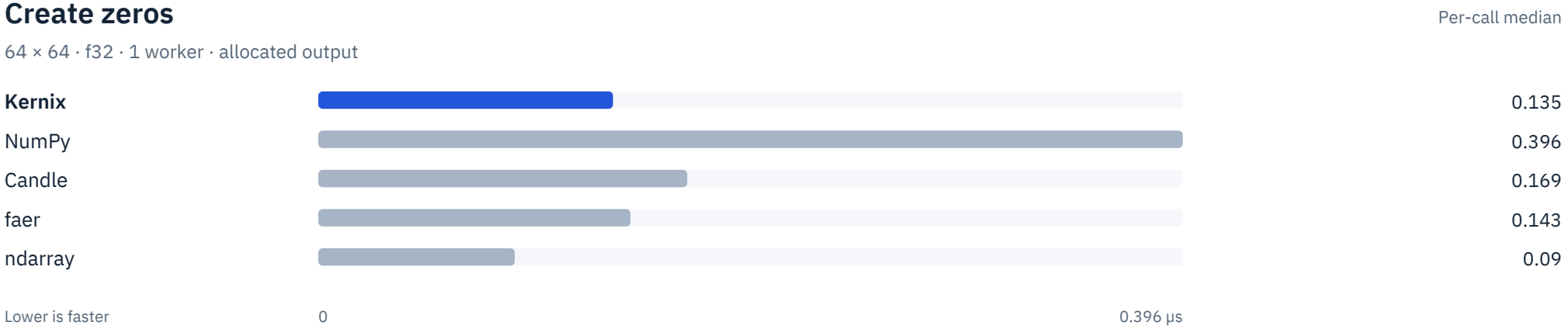


Kernix 0.7.2 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1589	1.589
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1412	1.412
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1498	1.498
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1875	1.875
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1500	1.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2822	2.822
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9138	9.138
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11253	11.253
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11299	11.299
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10916	10.916

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9862	9.862
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	13142	13.142
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	73035	73.035
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	119839	119.839
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	118439	118.439
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	86168	86.168
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	73442	73.442
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	79676	79.676
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	572709	572.709
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	702672	702.672

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	682415	682.415
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	742204	742.204
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	579849	579.849
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	608764	608.764
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5884	5.884
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5847	5.847
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6161	6.161
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5350	5.35
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4779	4.779
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6338	6.338

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	49181	49.181
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51443	51.443
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53199	53.199
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	41712	41.712
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	34791	34.791
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39220	39.22
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	393444	393.444
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	391705	391.705
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	398043	398.043
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	312433	312.433

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	278711	278.711
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	286706	286.706
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1936.0000000000002	1.9360000000000002
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2078	2.078
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3332	3.332
Thin QR / Kernix 96 × 32 · f64	42746	42.746
Thin QR / nalgebra 96 × 32 · f64	37668	37.668
Thin QR / faer 96 × 32 · f64	64468.99999999999	64.46899999999999
Thin QR / NumPy 96 × 32 · f64	38141	38.141
Thin QR / Kernix 256 × 64 · f64	496680	496.68

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	371777	371.777
Thin QR / faer 256 × 64 · f64	407851	407.851
Thin QR / NumPy 256 × 64 · f64	348219	348.219
Cholesky factorization / Kernix 64 × 64 · f64	22798	22.798
Cholesky factorization / nalgebra 64 × 64 · f64	13105	13.105
Cholesky factorization / faer 64 × 64 · f64	8175.999999999999	8.175999999999999
Cholesky factorization / NumPy 64 × 64 · f64	10755	10.755
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2423	2.423
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4231	4.231
Model inference / Kernix batched normalized/sliced MLP · f64	2001	2.001

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3486	3.486
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2776	2.776
Model recording + forward / Candle batched normalized/sliced MLP · f64	5198	5.198
Model backward / Kernix prepared graph → fresh gradients · f64	3194	3.194
Model backward / Candle prepared graph → fresh gradients · f64	14451	14.451
Parameter update / Kernix transactional SGD, fixed gradients · f64	514	0.514
Row softmax / Kernix 256 × 64 · f64	85344	85.344
Row softmax / ndarray 256 × 64 · f64	71614	71.614
Row softmax / Candle 256 × 64 · f64	80122	80.122
Row softmax / NumPy 256 × 64 · f64	99838	99.838

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	135	0.135
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create zeros / faer 64 × 64 · f32	143	0.143
Create zeros / Candle 64 × 64 · f32	169	0.169
Create zeros / NumPy 64 × 64 · f32	396	0.396
Create ones / Kernix 64 × 64 · f32	201	0.201
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / faer 64 × 64 · f32	202	0.202
Create ones / Candle 64 × 64 · f32	237	0.237
Create ones / NumPy 64 × 64 · f32	1345	1.345